

PawTrace Feature Prioritization Using the MoSCoW Method

Priority	Feature	Reason
M — Must have	Real-time GPS location	Without location tracking, the product cannot solve the main problem of helping owners find lost pets.
M — Must have	Electronic fence / safe-area alert	This is key for loss prevention. If the pet leaves the safe zone, the owner must receive an alert.
M — Must have	NFC pet identity card	If someone finds the pet, they need a fast way to see the pet's information and contact the owner.
M — Must have	Basic mobile app map page	Owners need a simple app interface to check location, set safe zones, and receive alerts.
M — Must have	Pet profile and owner contact information	This supports both the app and NFC rescue flow. Without it, the finder cannot contact the owner.
S — Should have	Health monitoring: heart rate, body temperature, activity level	Important for daily pet care, but the product can still be demonstrated with location and identity features first.
S — Should have	Health dashboard in the app	Users should be able to understand health data clearly, but it is not as launch-blocking as GPS and NFC.
S — Should have	Movement history	Useful for understanding pet behavior and past routes, but real-time location is more important.
S — Should have	Clear push notifications	Alerts are important, but the first demo can still show alert logic even if notification design is basic.
C — Could have	AI-assisted health photo check	This is a useful bonus feature, but it is not the core of the product's safety function.
C — Could have	Vaccination record display	Helpful for pet identity and health records, but not essential for the first working prototype.
C — Could have	Pet traits/personality description	Helps finders understand the pet, but it is not required for the main rescue flow.
C — Could have	Advanced navigation to the pet	Basic map location is enough for the first version; detailed route guidance can come later.
W — Won't have this semester	Full medical diagnosis	The product should not replace a vet. AI results should only be basic suggestions.

Priority	Feature	Reason
W — Won't have this semester	Support for campus stray animal management system	This is a possible future direction, but the first focus is household pet owners.
W — Won't have this semester	Social features for pet owners	Pet socialization is not part of the core safety and health problem.
W — Won't have this semester	Advanced multi-pet organization platform	Useful later, but too large for the current semester scope.